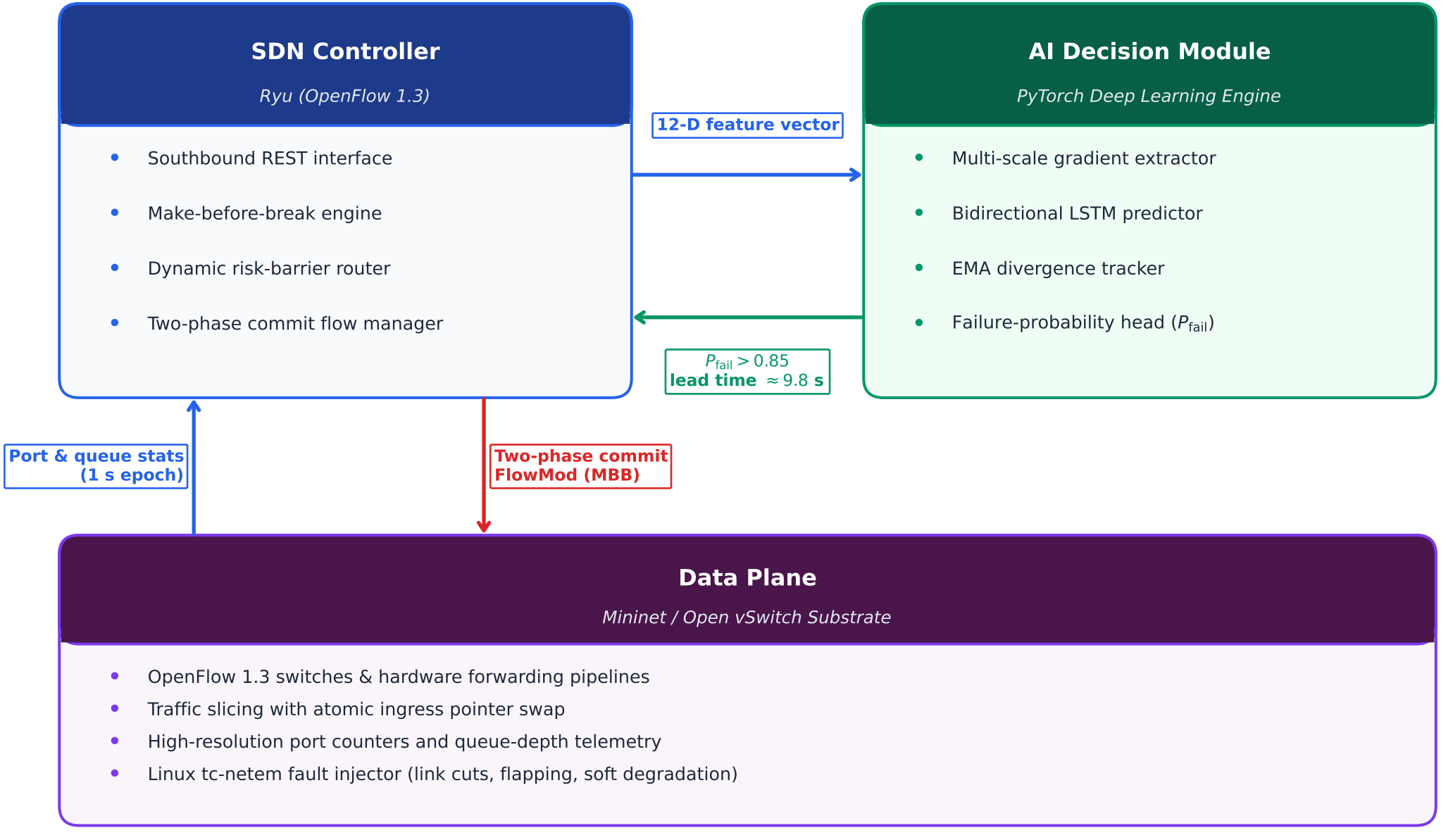
SDN Controller
Ryu (OpenFlow 1.3)
Southbound REST interface
Make-before-break engine
Dynamic risk-barrier router
Two-phase commit flow manager
12-D feature vector
AI Decision Module
PyTorch Deep Learning Engine
Multi-scale gradient extractor
Bidirectional LSTM predictor
EMA divergence tracker
Failure-probability head (P_{fail})
$P_{fail} > 0.85$
lead time ≈ 9.8 s
Port & queue stats
(1 s epoch)
Two-phase commit
FlowMod (MBB)
Data Plane
Mininet / Open vSwitch Substrate
OpenFlow 1.3 switches & hardware forwarding pipelines
Traffic slicing with atomic ingress pointer swap
High-resolution port counters and queue-depth telemetry
Linux tc-netem fault injector (link cuts, flapping, soft degradation)